Department of Engineering Science
School of Science & Technology

CES 552 Network Architecture
Midterm Review

Dr. Jingxian Wu
Jingxian.wu@sonoma.edu

IP

- **Internet Protocol Header**
  - Fragmentation: Id, DF, MF, Frag. Offset
- **IP address**
  - Binary, hexadecimal, dotted decimal
  - Class A, B, C, special (this host, broadcast, loopback)
  - Subnet, subnet mask
  - NAT (example during lecture)
- **ICP**
  - Carried in payload field of IP packet
  - ICMP (echo request, echo reply)
  - ARP, DHCP

IP

- **Routing in Internet**
  - Interior Gateway Protocol (IGP): OSPF (Link State Routing)
  - Boarder Gateway Protocol (BGP): enhanced distance vector routing
- **IPv6**
  - 128 bits
  - Address notation (eight groups of four hexadecimal digits)

TRANSPORT LAYER

- **Functions & Serverce**
  - true end-to-end
  - Connection oriented v.s. connectionless in transport layer
  - Service primitives:
    - Definition
    - Typical primitives: listen, connect, send, receive, disconnect
    - What is socket?
- **Elements of transport protocol**
  - Addressing: port
  - Connection establishment: 3-way hand-shake
  - Connection release: asymmetric v.s. symmetric
  - Flow control: sliding window with variable window size
  - Crash recovery
    - Special occasion that crash cannot be recovered at transport layer.

TRANSPORT LAYER

- **UDP**
  - Unreliable Connectionless
  - Why UDP?
  - Example applications: DHCP, DNS, RTP
  - Properties of RTP:
    - Provide basic functions required by most real-time applications, no flow control, doesn't provide QoS
- **TCP**
  - Reliable connection-oriented
  - Header:
    - ACK, RST, SYN, FIN, Ack#, Window size
    - Options: window scale option, select repeat of go back N
  - Connection Establishment
    - Three way hand-shake (packet header fields)
  - Connection Release
    - Symmetric release (packet header fields)

TRANSPORT LAYER

- **TCP (Cont'd)**
  - Flow control (variable window size)
  - Congestion control
    - Different between congestion control and flow control
    - Transmitter window size calculation
  - Timer management
    - Retransmission timer calculation
    - Persistence timer, keepalive timer
- **Performance**
  - Metrics (throughput, delay, jitter, packet loss rate)
  - Bandwidth, throughput, goodput
  - Bandwidth-delay product

APPLICATION LAYER

- **DNS**
  - On top of UDP
  - Hierarchical structure (top-level domain, sub-domain)
  - Distributed
  - Resource records
    - A, MX, NS, CNAME